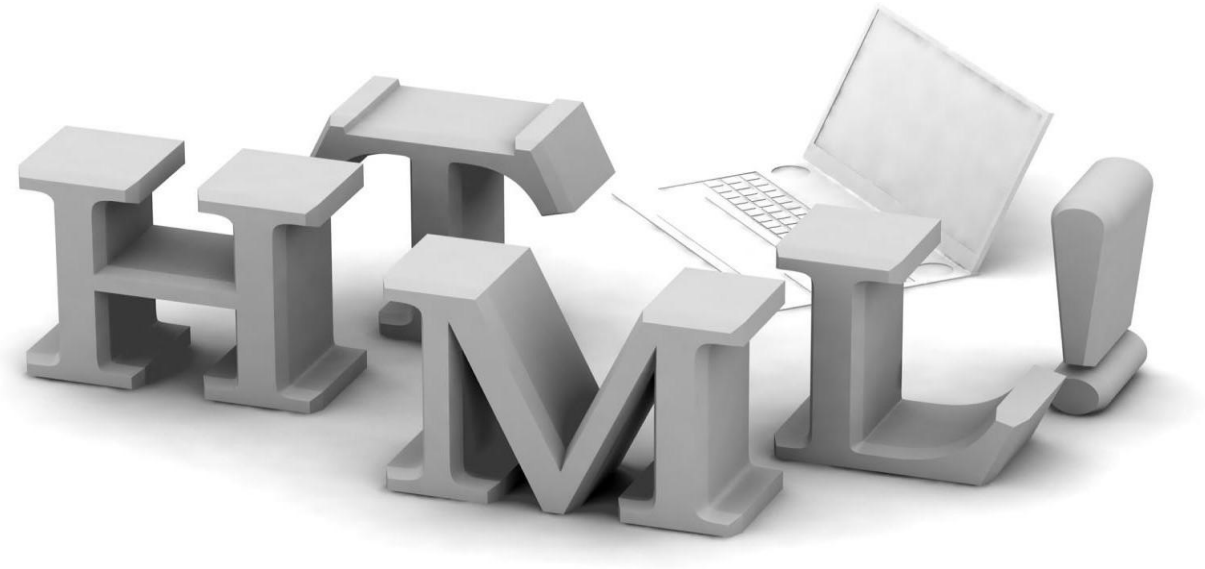
Faculty of Computer Science and Mathematics

Lab Exercise 2

Name:	AHMAD ISA ISKANDAR BIN CHOI SUNG BOK
Matric Num:	S80787



Overview:

This in-class exercise extends Lab Exercise 2. You will apply tables (including colspan and rowspan), forms (text inputs, radio buttons, checkboxes, drop-down lists and labels) and semantic HTML tags (header, nav, section, article and footer) to solve a realistic problem.

There are two problems. Your lecturer will assign one problem to you, or you may be asked to complete both.

Learning Outcomes

1. Create tables with headers and merged cells using colspan and rowspan.

2. Build forms with appropriate input types, correctly linked labels and grouped radio buttons.

3. Structure a web page using semantic HTML tags.

4. Identify and correct errors in tables, forms and semantic structure, and validate code using the W3C Validator.

Best of luck!

Problem A: Faculty Event Page with Schedule and Registration

Your faculty is organising a one-day event, for example a Programming Workshop, a Career Day or a Hackathon. The organising committee needs a simple web page that shows the event schedule and allows students to register. Build the page using HTML only (no CSS). You may choose any event name, date and activities.

Instructions

Create a file named lab2_ExerA.html that contains the following:

1. Semantic Structure

- A `<header>` containing an `<h1>` with the event name and a short tagline.
- A `<nav>` with internal links (e.g. `href="#schedule"`, `href="#register"`, `href="#about"`) that jump to each section of the page.
- At least three `<section>` elements, each with an `id`: About the Event, Event Schedule and Registration.
- Inside the About section, two `<article>` elements, each introducing one speaker or one activity (with an `<h3>` and a `<p>`).
- A `<footer>` with the organiser name and a copyright line using `©`.

2. Event Schedule Table

- A table with `border="1"` and header cells (`<th>`) for Time, Activity, Venue and Person in Charge.
- At least 6 rows of activities from morning to evening.
- Use `colspan` at least once (e.g. a "Lunch Break" row that spans across Activity, Venue and Person in Charge).
- Use `rowspan` at least once (e.g. two consecutive sessions held in the same venue).

3. Registration Form

Create a form with `action="register.html"` and `method="post"` that contains:

- Text inputs for Full Name, Matric Number and E-mail, each with a `<label>` whose `for` matches the input `id`.
- Radio buttons for Year of Study (Year 1 – Year 4). All radio buttons must share the same `name` so only one can be selected.
- Checkboxes for the sessions the student wants to join (at least 3 choices).
- A drop-down list (`<select>`) for Programme or Faculty with at least 4 options.
- A submit button.

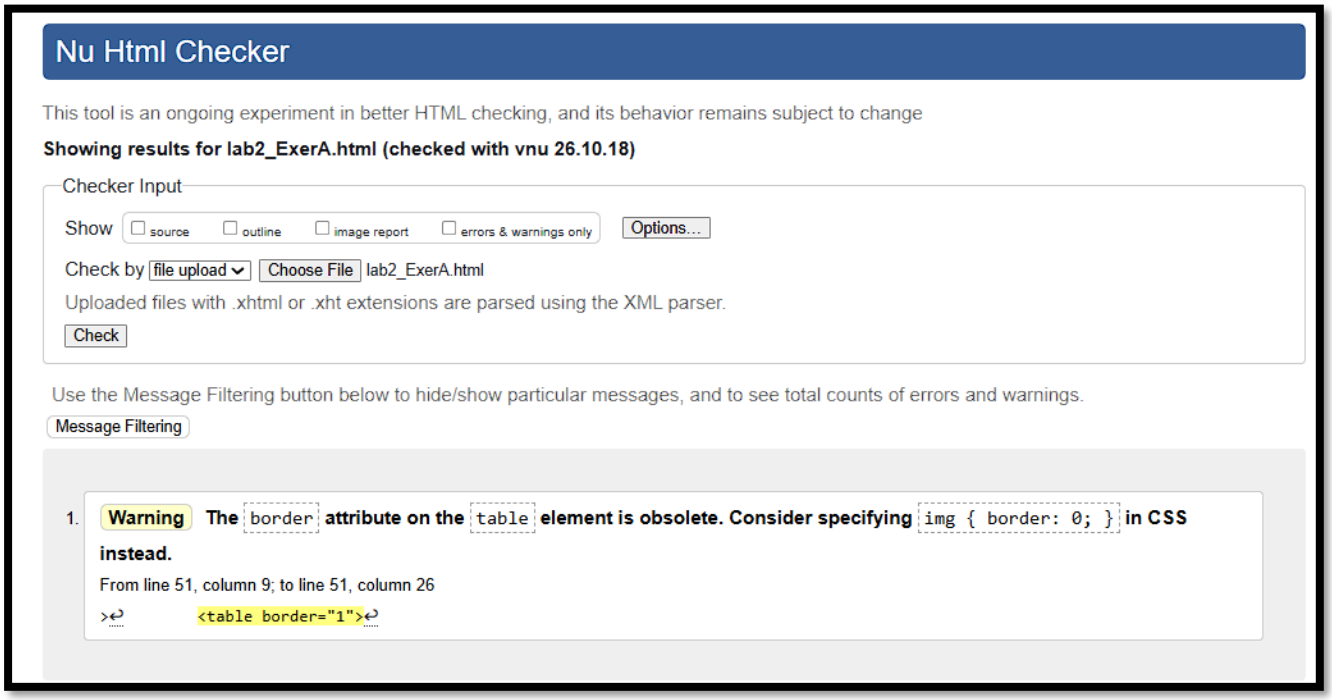
4. Comments

- Use comments (`<!-- ... -->`) to explain each section of your code.

Submission

Save the file as lab2_ExerA.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab2_ExerA.html and Screenshot of validation) through Github

VALIDATION:


Nu Html Checker
This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change
Showing results for lab2_ExerA.html (checked with vnu 26.10.18)
Checker Input
Show
source
outline
image report
errors & warnings only
Options...
Check by
file upload
Choose File
lab2_ExerA.html
Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.
Check
Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.
Message Filtering
1. Warning The border attribute on the table element is obsolete. Consider specifying img { border: 0; } in CSS instead.
From line 51, column 9; to line 51, column 26
><table border="1">↩

Code(lab2_ExerA.html):

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Programming Workshop</title>
</head>
<body>
    <!-- Header section -->
    <header>
        <h1>Programming Workshop</h1>
        <p>Join us for an exciting programming workshop
            where you can enhance your coding skills and learn
            from industry experts.</p>
    </header><br>

    <!-- Navigation for three tags -->
    <nav>
        <!-- Using unordered list -->
        <h2>Navigation</h2>

        <a href="#schedule">Schedule</a> |
        <a href="register.html">Registration</a> |
        <a href="#about">About</a>

    </nav><br>

    <!-- Section id set as 'schedule' to link the hyperlink at nav -->
    <section id="about">
        <h2>About the Workshop</h2>
        <p>The workshop will be held over two days, with sessions covering various
            programming languages and techniques.</p>
```

```
67          <tr>
68              <td>10:30 AM - 12:00 PM</td>
69              <td>Java Basics</td>
70              <td>BYOD 2</td>
71              <td>Dr ts. Amier</td>
72          </tr>
73
74          <tr>
75              <td >12:00 PM - 1:00 PM</td>
76              <!-- Using colspan to span across multiple columns-->
77              <td colspan="3" style="text-align: center;">Lunch Break</td>
78          </tr>
79
80          <tr>
81              <td>1:00 PM - 3:00 PM</td>
82              <!-- Using rowspan to span 2 rows-->
83              <td rowspan="2">Hands-on Java Coding</td>
84              <td>BYOD 1</td>
85              <td>Encik Mal</td>
86          </tr>
87
88          <tr>
89              <td>3:00 PM - 4:30 PM</td>
90              <td>BYOD 2</td>
91              <td>Encik Abu</td>
92          </tr>
```

```
34          <!-- Using article as a block of content -->
35          <article>
36              <h3>Speaker: Dr John</h3>
37              <p>Dr John is a renowned software engineer with over 15 years of experience
38                  in the industry. He will be sharing his insights on modern programming practices.</p>
39          </article>
40
41          <article>
42              <h3>Activity: Java Coding</h3>
43              <p>Participants will engage in hands-on Java coding exercises,
44                  allowing them to apply what they have learned in a practical setting.</p>
45          </article>
46      </section>
47
48      <!-- Section for Event Schedule-->
49      <section id="schedule">
50          <h2>Event Schedule Table</h2>
51          <table border="1">
52              <!-- Using table row, table header and table data-->
53              <tr>
54                  <th>Time</th>
55                  <th>Activity</th>
56                  <th>Venue</th>
57                  <th>Person In Charge</th>
58              </tr>
59
60              <tr>
61                  <td>9:00 AM - 10:30 AM</td>
62                  <td>Introduction to Programming</td>
63                  <td>BYOD 1</td>
64                  <td>Dr Haikal</td>
65              </tr>
```

```
89                  <td>3:00 PM - 4:30 PM</td>
90                  <td>BYOD 2</td>
91                  <td>Encik Abu</td>
92              </tr>
93
94              <tr>
95                  <td>4:30 PM - 5:00 PM</td>
96                  <td>Q&A Session</td>
97                  <td>BYOD 1</td>
98                  <td>Prof Dr Madya ts. Mad</td>
99              </tr>
100         </table>
101     </section>
102 </body>
103 </html>
```

Code (register.html)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Registration Form for Programming Workshop</title>
</head>
<body>
    <section>
        <h2>Registration</h2>
        <!-- Using form element and id register.html to set the hyperlink from homepage-->
        <form  id="register.html" method="post" action="lab2_ExerA.html">
            <!-- Using type text for textbox -->
            <label for="name">Full Name:</label>
            <input type="text" id="name" name="name"><br><br>

            <label for="matric">Matric Number:</label>
            <input type="text" id="matric" name="matric"><br><br>

            <!-- Using type email for email input that contains @ -->
            <label for="email">Email:</label>
            <input type="email" id="email" name="email"><br><br>

            <!-- Using radio buttons for year of study -->
            <label for="year">Year of Study:</label><br>
            <input type="radio" id="year1" name="year" value="1">
            <label for="year1">Year 1</label>

            <input type="radio" id="year2" name="year" value="2">
            <label for="year2">Year 2</label>

            <input type="radio" id="year3" name="year" value="3">
            <label for="year3">Year 3</label>

            <input type="radio" id="year4" name="year" value="4">
            <label for="year4">Year 4</label><br><br>
```

```
38              <!-- Select Sessions  using checkboxes-->
39              <label for="sessions">Select Sessions:</label><br>
40              <input type="checkbox" id="session1" name="sessions" value="Session 1">
41              <label for="session1">Session 1: Introduction to Programming</label><br>
42
43              <input type="checkbox" id="session2" name="sessions" value="Session 2">
44              <label for="session2">Session 2: Advanced Programming Techniques</label><br>
45
46              <input type="checkbox" id="session3" name="sessions" value="Session 3">
47              <label for="session3">Session 3: Hands-on Coding Exercises</label><br><br>
48
49              <!-- Using drop down list -->
50              <label for="program">Program of Study:</label>
51              <select id="program" name="program">
52                  <option value="Computer Science">Computer Science</option>
53                  <option value="Software Engineering">Software Engineering</option>
54                  <option value="Information Technology">Information Technology</option>
55                  <option value="Data Science">Science Marine</option>
56              </select><br><br>
57
58              <input type="submit" value="Register">
59          </form>
60      </section>
61  </body>
62  </html>
```

Problem B: Course Feedback Page (Debugging and Extend)

A student helper created a Course Feedback page for CSF3133. The page shows a feedback summary table and a feedback form, but it contains several errors. Some radio buttons can be selected at the same time, clicking some labels does nothing, and the table looks broken. Your tasks are to find and fix the errors, then extend the page.

Given Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <title>Course Feedback</title>
</head>
<body>
<header>
    <h1>CSF3133 Course Feedback</h1>
</head>
<nav>
    <a href="#results">Results</a> | <a href="#form">Feedback Form</a>
</nav>
<section id="results">
    <h2>Feedback Summary</h2>
    <table border="1">
        <tr>
            <th>Week</th>
            <th>Topic</th>
            <td>Rating</td>
        </tr>
        <tr>
            <td>1</td>
            <td>Introduction to HTML</td>
            <td>4.5</td>
        <tr>
            <td>2</td>
            <td colspan="3">Tables and Forms</td>
        </tr>
        <tr>
            <td>3</td>
            <td>Semantic HTML<td>4.0</td>
        </tr>
    </table>
</section>
<section id="form">
    <h2>Give Your Feedback</h2>
    <form action="submit_feedback.php" method="post">
        <label for="name">Name:</label>
        <input type="text" id="fullname" name="name"><br>
        <label>Overall rating:</label><br>
        <input type="radio" id="good" name="good" value="good">
        <label for="good">Good</label>
        <input type="radio" id="poor" name="poor" value="poor">
        <label for="poor">Poor</label><br>
        <input type="checkbox" id="lab" name="lab" value="yes">
        <label for="labs">I attended all labs</label><br>
        <input type="submit" value="Submit">
    </from>
</section>
<footer>
    <p>Copyright &copy 2026 CSF3133</p>
</footer>
</body>
</html>
```

Instructions

Part 1. Find and Fix the Errors

1. Copy the given code into a new file named lab2_ExerB.html.

2. Identify and correct all errors. At the top of the `<body>`, write an HTML comment listing each error and the line number, e.g. `<!-- Error 1 (Line 1): missing <!DOCTYPE html> -->`.

4. Validate your corrected file at `https://validator.w3.org`. Your final page must show "No errors or warnings to show".

Part 2. Extend into a Two-Page Website

5. Table: add rows for Week 4 and Week 5. Both weeks cover the same topic "Mini Project", so use `rowspan="2"` on the Topic cell. Add a final row "Average Rating" that uses `colspan="2".`

6. Form: add an E-mail text input with a matching `<label>`, a third radio option "Average", a group of at least 3 checkboxes "Topics I want more practice on", and a `<select>` for Section/Group (at least 3 options).

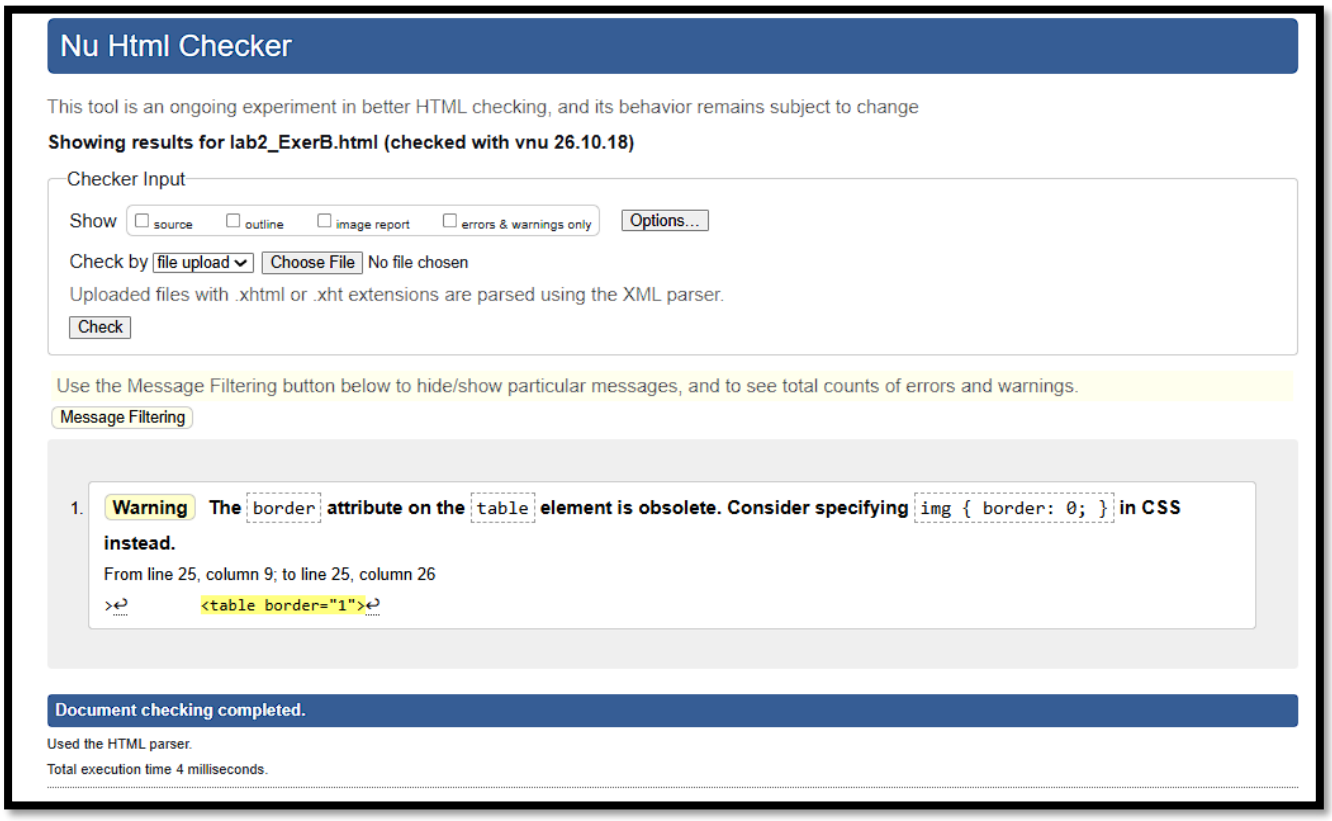
7. Semantic HTML: add a new `<section id="comments">` titled "Student Comments" that contains two <article> elements, each with an `<h3>` (student name) and a `<p>` (comment). Add a link to this section in the `<nav>`.

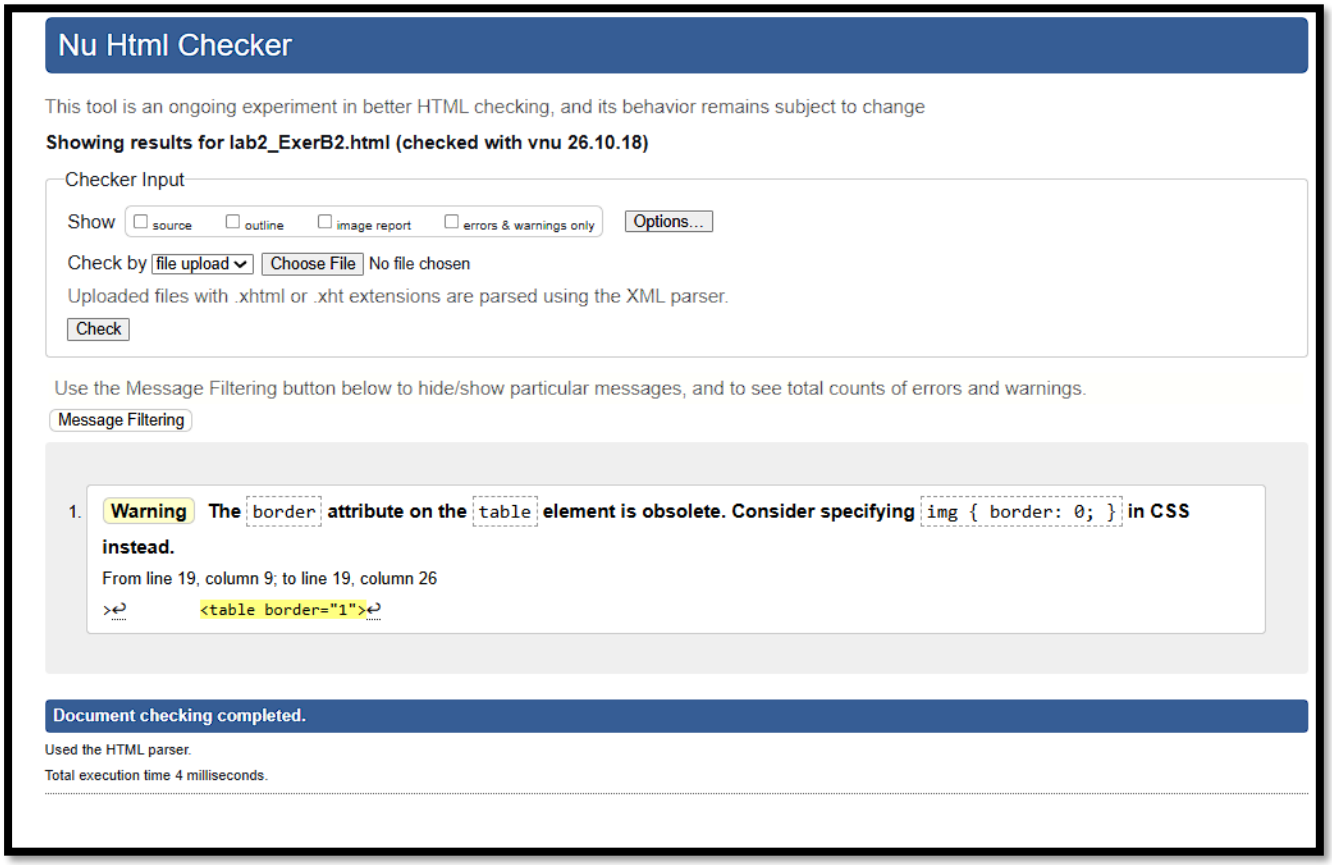
8. Use comments to explain each new section of your code.

Submission

Save the file as lab2_ExerB.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab2_ExerB.html and Screenshot of validation) through Github

Validation:


Nu Html Checker
This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change
Showing results for lab2_ExerB.html (checked with vnu 26.10.18)
Checker Input
Show source outline image report errors & warnings only Options...
Check by file upload Choose File No file chosen
Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.
Check
Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.
Message Filtering
1. Warning The border attribute on the table element is obsolete. Consider specifying img { border: 0; } in CSS instead.
From line 25, column 9; to line 25, column 26
><table border="1">
Document checking completed.
Used the HTML parser.
Total execution time 4 milliseconds.


Nu Html Checker
This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change
Showing results for lab2_ExerB2.html (checked with vnu 26.10.18)
Checker Input
Show source outline image report errors & warnings only Options...
Check by file upload Choose File No file chosen
Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.
Check
Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.
Message Filtering
1. Warning The border attribute on the table element is obsolete. Consider specifying img { border: 0; } in CSS instead.
From line 19, column 9; to line 19, column 26
><table border="1">
Document checking completed.
Used the HTML parser.
Total execution time 4 milliseconds.

Code (lab2_ExerB.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <title>Course Feedback</title>
</head>
<body>
    <!--
      Error 1 (Line 10): Incorrect closing tag </head> instead of </header>
      Error 2 (Line 28): colspan="3" exceeds the 3-column table limit; changed to colspan="2"
      Error 3 (Line 32): Typo in cell separation; replaced inner <td> with </td>
      Error 4 (Line 40): Input id="fullname" does not match label for="name"; changed id to "name"
      Error 5 (Lines 42 & 44): Radio buttons have different names; grouped both under name="rating"
      Error 6 (Line 47): Label for="labs" does not match input id="lab"; changed for to "lab"
      Error 7 (Line 49): Typo in closing tag </from> instead of </form>
    -->
    <header>
        <h1>CSF3133 Course Feedback</h1>
    </header>
    <nav>
        <a href="#results">Results</a> | <a href="#form">Feedback Form</a>
    </nav>
    <section id="results">
        <h2>Feedback Summary</h2>
        <table border="1">
            <tr>
                <th>Week</th>
                <th>Topic</th>
                <th>Rating</th>
            </tr>
            <tr>
                <td>1</td>
                <td>Introduction to HTML</td>
                <td>4.5</td>
            </tr>
```

```
            </tr>
            <tr>
                <td>2</td>
                <td colspan="2">Tables and Forms</td>
            </tr>
            <tr>
                <td>3</td>
                <td>Semantic HTML</td>
                <td>4.0</td>
            </tr>
        </table>
    </section>
    <section id="form">
        <h2>Give Your Feedback</h2>
        <form action="submit_feedback.php" method="post">
            <label for="name">Name:</label>
            <input type="text" id="name" name="name"><br>

            <label>Overall rating:</label><br>
            <input type="radio" id="good" name="rating" value="good">
            <label for="good">Good</label>
            <input type="radio" id="poor" name="rating" value="poor">
            <label for="poor">Poor</label><br>

            <input type="checkbox" id="lab" name="lab" value="yes">
            <label for="lab">I attended all labs</label><br>

            <input type="submit" value="Submit">
        </form>
    </section>
```

```
    <footer>
        <p>Copyright &copy; 2026 CSF3133</p>
        <h2>Part 2: Extend into a Two-Page Website</h2>
        <a href="lab2_ExerB2.html">Click Here For Part 2</a>
    </footer>
</body>
</html>

```

Code (lab2_ExerB2.html)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <title>Course Feedback</title>
</head>
<body>
    <header>
        <h1>CSF3133 Course Feedback</h1>
    </header>
    <nav>
        <a href="#results">Results</a> |
        <a href="#form">Feedback Form</a> |
        <!-- Added navigation link to the student comments section -->
        <a href="#comments">Student Comments</a>
    </nav>
    <section id="results">
        <h2>Feedback Summary</h2>
        <table border="1">
            <tr>
                <th>Week</th>
                <th>Topic</th>
                <th>Rating</th>
            </tr>
            <tr>
                <td>1</td>
                <td>Introduction to HTML</td>
                <td>4.5</td>
            </tr>
            <tr>
                <td>2</td>
                <td colspan="2">Tables and Forms</td>
            </tr>
            <tr>
```

```
            <tr>
                <td>3</td>
                <td>Semantic HTML</td>
                <td>4.0</td>
            </tr>
            <!-- Added rows for Week 4 & 5 using rowspan="2" on the shared Topic cell -->
            <t
                The tr element represents a row of cells in a table.
                Widely available across major browsers (Baseline since 2015)
                MDN Reference
            </tr>
            <tr>
                <td>5</td>
                <td>4.7</td>
            </tr>
            <!-- Added final summary row using colspan="2" -->
            <tr>
                <td colspan="2">Average Rating</td>
                <td>4.35</td>
            </tr>
        </table>
    </section>
    <section id="form">
        <h2>Give Your Feedback</h2>
        <form action="lab2_ExerB2.html" method="post">
            <label for="name">Name:</label>
            <input type="text" id="name" name="name"><br>

            <!-- Added E-mail input with matching label -->
            <label for="email">E-mail:</label>
            <input type="email" id="email" name="email"><br>

            <label>Overall rating:</label><br>
            <input type="radio" id="good" name="rating" value="good">
            <label for="good">Good</label>
            <!-- Added third radio button option "Average" -->
            <input type="radio" id="average" name="rating" value="average">
            <label for="average">Average</label>
            <input type="radio" id="poor" name="rating" value="poor">
            <label for="poor">Poor</label><br>

            <!-- Added section dropdown selection -->
            <label for="section">Section/Group:</label>
            <select id="section" name="section">
                <option value="group1">Group 1</option>
                <option value="group2">Group 2</option>
                <option value="group3">Group 3</option>
            </select><br>

            <!-- Added practice checkboxes group -->
            <label>Topics I want more practice on:</label><br>
            <input type="checkbox" id="practice_html" name="practice[]" value="html">
            <label for="practice_html">HTML Basics</label><br>
            <input type="checkbox" id="practice_tables" name="practice[]" value="tables">
            <label for="practice_tables">Tables</label><br>
            <input type="checkbox" id="practice_forms" name="practice[]" value="forms">
            <label for="practice_forms">Forms</label><br>
```

```
            <input type="checkbox" id="lab" name="lab" value="yes">
            <label for="lab">I attended all labs</label><br>

            <input type="submit" value="Submit">
        </form>
    </section>

    <!-- Added Student Comments Section with two article components -->
    <section id="comments">
        <h2>Student Comments</h2>
        <article>
            <h3>Alex Tan</h3>
            <p>The practical labs were highly engaging and helped solidify my core layout concepts quickly.</p>
        </article>
        <article>
            <h3>Sarah Jenkins</h3>
            <p>Form elements validation was tricky at first, but the course exercises made it much clearer.</p>
        </article>
    </section>

    <footer>
        <p>Copyright &copy; 2026 CSF3133</p>
    </footer>
</body>
</html>

```

Rubric

Table A: Rubric for Additional Lab Exercise 2 – Problem A

Criteria	Description	Marks
Table Creation	Schedule table has <th> headers, at least 6 rows of meaningful data, and correct use of colspan and rowspan (no extra or missing cells).	3
Form Structure	Form contains text inputs with matching <label for> and id, a grouped set of radio buttons (same name), at least 3 checkboxes, a <select> with at least 4 options, and a submit button.	3
Semantic HTML	Proper use of <header>, <nav> with working internal links, <section> (min. 3), <article> (min. 2) and <footer>.	2
Code Organization and Readability	Code is well-organized, properly indented and commented. File is named lab2a.html and passes the W3C Validator.	2
Total		**10**

Table B: Rubric for Additional Lab Exercise 2 – Problem B

Criteria	Description	Marks
Error Identification	At least 7 errors correctly identified with line numbers and listed in an HTML comment.	3
Error Correction & Validation	All errors corrected. Radio buttons work as one group, all labels are linked, and the page passes the W3C Validator.	2
Table Extension	Week 4 and Week 5 rows added with correct rowspan; Average Rating row uses correct colspan. No extra or missing cells.	2
Form Extension	E-mail input with label, third radio option in the same group, at least 3 checkboxes and a <select> with at least 3 options.	2
Semantic HTML and Code Organization	New <section> with two <article> elements and a working <nav> link. Code is indented and commented.	1
Total		**10**